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Is bitcoin an inflation hedge?

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ABSTRACT

Spot bitcoin ETFs have been recently approved in the U.S., increasing retail and institutional investors' attention to crypto. To contribute to the debate on whether bitcoin protects against inflation, we analyze the effect of inflation shocks on bitcoin returns through the estimation and inference of Vector Autoregressive Models (VARs), identifying inflation shocks as surprises in the U.S.'s CPI and Core PCE announcements. Based on monthly data between August 2010 and January 2023, the results indicate that bitcoin returns increase significantly after a positive inflationary shock, corroborating empirical evidence that bitcoin can act as an inflation hedge. However, we observe that bitcoin's inflationary hedging property is sensitive to the price index – it only holds for CPI shocks – and to the period of analysis – the hedging property stems primarily from sample periods before the increasing institutional adoption of BTC ("early days"). Notably, the inflation hedge property of bitcoin (Gold) has disappeared (strengthened) from the COVID-19 outbreak onwards. We conclude that bitcoin's inflation-hedging property is context-specific and likely diminishes as it achieves broader adoption and becomes more integrated into mainstream financial markets.

1. Introduction

Inflation is a critical investment risk and one of the foremost challenges to conducting business in the upcoming years (World Economic Forum, 2023). Consequently, investors are urged to formulate strategies to mitigate inflation risk. A viable safeguard involves incorporating in the portfolio assets that appreciate alongside inflation (Tarbert, 1996). Such a need has gained even more importance given the recent generalized price increase in various parts of the world, including developed economies, where low interest rates and controlled inflation have coexisted for decades. But not anymore.

The recent consideration of Bitcoin as an investment instrument with considerable portfolio implications (Guesmi et al., 2019; Conlon et al., 2021), along with the growing interest among academics and policymakers (Phochanachan et al., 2022), has contributed to positioning the cryptocurrency as a potential inflation hedge. Commonly cited justifications for this role include its limited supply and decentralized network, conferring scarcity and resilience (Bouri et al., 2017b,a). However, existing theoretical and empirical studies have not reached a consensus on Bitcoin's ability to hedge for inflation – while Blau et al. (2021) and Choi and Shin (2021) support Bitcoin as a robust inflation hedge, most studies indicate that Bitcoin inflation hedging properties are context-specific (Matkovskyy & Jalan, 2020; Conlon et al., 2021; Smales, 2022; Phochanachan et al., 2022; Gaies et al., 2024), including bitcoin futures (Liu & Valcarcel, 2024). This lack of consensus may be attributed to methodological differences, varying time horizons, and the specific characteristics of the economies analyzed in these studies.

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One key aspect of answering this research question is whether a given inflation announcement carries *unexpected* information or not. To demonstrate this point, let us consider the following two situations. If the observed CPI matches analysts' consensus (i.e., market expectations), asset prices should not react, regardless of the level of inflation. However, if at least part of the information was a *surprise*, the announcement generates new information for financial markets, and asset prices should move accordingly. Based on this concept, Smales (2022) finds that bitcoin returns are negatively related to inflation surprises (proxied by the CPI) on the announcement day. Meanwhile, outside a contemporaneous relationship, empirical research that models the dynamic relationship between cryptocurrency returns and innovations in inflation (e.g. Choi & Shin, 2021; Liu & Valcarcel, 2024), to our knowledge, has relied solely on expected or observed inflation.

To contribute to the inconclusive but critical debate on the inflation-hedging properties of cryptocurrency, we estimate and analyze the short-term effect of inflation surprises on Bitcoin price fluctuations through VARs. Selecting a VAR framework is justified because structural analyses (such as impulse-response analysis and forecast error variance decomposition) enable us to assess the dynamic interactions and feedback mechanisms among the variables and examine how persistent the inflation effects are.[1] Particularly, instead of identifying inflation shocks as the residual from a typical regression of observed (or expected) inflation on the lags of itself and of several variables (S&P 500, VIX, interest rates, etc.) (for example, as in Choi & Shin, 2021), we focus on the unexpected component of inflation releases, measured here as the difference between actual monthly inflation and the consensus forecast of market analysts. By doing that, our innovations (or shocks) consider all observed and unobserved factors that market agents consider in their projections of inflation, and not only a few covariates included in the VAR system.[2] Following previous studies, we include in the system of equations the joint effects of gold prices, the S&P500 and VIX indexes, the one-year US Treasury bill yield, and bitcoin prices. Our data is monthly and ranges from August 2010 to January 2023.

The results of the impulse response functions indicate that Bitcoin returns increase following a positive inflation shock, ceteris paribus. Compared to other assets, this pattern is similar to the responses observed in gold and contrasts with the one observed in the S&P 500. Further analyses reveal that corporate bonds follow a similar pattern to the S&P 500, while real estate aligns with bitcoin and gold. While these findings suggest that Bitcoin may act as a hedge against unexpected fluctuations in inflation, the variance decomposition analysis indicates that the return-inflation relationship exhibits greater strength for gold than for Bitcoin, consistent with the findings of Smales (2022).

Further analyses suggest that the efficacy of Bitcoin as an inflation hedge appears contingent on two aspects: the selected price index and the sample period. Regarding the former, when considering the Core PCE index instead of the CPI, the cumulative dynamic responses of bitcoin to a positive inflation shock became negative. This discrepancy may stem from different criteria used in the indices, such as composition, weighting, and the adaptation of expectations, as the PCE is released after the CPI. Regarding the former, we find a decrease in the inflation-hedging properties of Bitcoin when we exclude the initial sampling period ("early days" of Bitcoin) and a reversal when we consider the subperiod from COVID-19 onwards, suggesting that such hedging property may be diminishing as adoption – and, consequently, market fluctuations – become mainstream. Consistent with this argument, Nguyen (2022) finds that, during periods of high uncertainty, such as COVID-19, the S&P500 returns significantly affected bitcoin returns. As more investors adopt Bitcoin and trading participants resemble those in other markets such as equities, the inflation-hedging property of Bitcoin is likely to diminish.

This research contributes to a growing but still underexplored debate on Bitcoin hedging properties in the face of inflationary shocks (Blau et al., 2021; Choi & Shin, 2021; Matkovskyy & Jalan, 2020; Conlon et al., 2021; Smales, 2022; Phochanachan et al., 2022). Notably, our contributions are twofold. First, we show that the dynamic cumulative response of bitcoin returns is generally positive for surprises in the CPI (MoM and YoY) but negative for surprises in the Core PCE. Such a result points out the importance of the inflation index to address the research question. Second, in contrast to the few studies that considered unexpected inflation in their analyses (Matkovskyy & Jalan, 2020; Smales, 2022), our empirical methodology takes into account the dynamic and temporal relationships among the variables in the model, which is especially relevant in time series analyses. At the same time, we seek to mitigate possible endogeneity problems that often arise in this type of investigation.

Furthermore, as this is a current discussion with significant practical implications for asset managers, investors, and monetary authorities, the empirical evidence presented in this study directly affects resource allocation decisions and allows for a recent analysis of the behavior of Bitcoin, gold, and other assets following inflation shocks. In particular, recent research has shown that sophisticated retail investors are attracted to cryptocurrencies for macroeconomic and portfolio reasons (Colombo & Yarovaya, 2024). Our study is also timely since the SEC approved spot Bitcoin ETFs only in 2024, bringing tremendous attention and demand for Bitcoin exposure in retail investors' portfolios.

The remainder of this paper is organized as follows: Section 2 presents the theoretical framework and methodology used in this study. Section 3 presents the data used in this work. Section 4 reports the empirical results, and Section 5 presents the conclusions.

[1] Using a similar empirical framework but focusing on monetary policy shocks, Ma et al. (2022) find that the response of Bitcoin prices is long-lasting and permanent.

[2] As Nakamura and Steinsson (2017) note, any factors left out as covariates in the regression will be part of the "shock", even if they are anticipated by market agents.

2. Methodology

2.1. VAR model

To capture the dynamics of the variables and deal with the potential endogeneity in their relationships, we use a VAR framework (Sims, 1980), which is reliable in describing, summarizing, and forecasting macroeconomic data (Stock & Watson, 2001).

Specifically, let $y_1, y_2, \ldots, y_T$ be multivariate time series, with $y_t = (y_{1t}, y_{2t}, \ldots, y_{kt})'$. A VAR model of order p, in its basic form, can be expressed as follows:

$$y_t = v + A_1 y_{t-1} + A_2 y_{t-2} + \ldots + A_p y_{t-p} + u_t \quad ; \quad t = 1, 2, 3, \ldots, N \tag{1}$$

All symbols used in this representation have usual meanings, that is, $y_t = (y_{1t}, y_{2t}, \ldots, y_{kt})'$ is a random vector $(K \times 1)$ of endogenous variables; $v = (v_1, v_2, \ldots, v_k)'$ is a fixed vector of intercepts $(K \times 1)$, which allows the possibility of a non-zero mean $E(y_t)$; A_i are fixed matrices of coefficients $(K \times K)$ where $i = 1, \ldots, p$, which are interpreted as the sensitivity of a model variable about a lag of another variable; $u_t = (u_{1t}, u_{2t}, \ldots, u_{kt})'$ is a K-dimensional vector of white noise, such that $E(u) = 0$, $E(u_t u') = \Sigma_u$, and $E(u_t u') = 0$ for $s \neq t$, that is, it represents random shocks that do not correlate and are time-invariant.

Under the stability condition, the process Y_t has the MA representation defined as:

$$Y_t = \mu + \sum_{i=0}^{\infty} A^i U_{t-i} \tag{2}$$

Where Y_t is modeled as a function of the mean term μ and in terms of the past and present of the innovation vector U_t.

Furthermore, the MA representation of y_t can be found by pre-multiplying Y_t by a matrix $J = [I_K: 0: \ldots: 0]$ of dimension $(K \times Kp)$. Here, $\mu = J\mu$, $\Phi_i = JA^iJ'$, and $u_t = JU_t$.

$$y_t = JY_t = J\mu + \sum_{i=0}^{\infty} \Phi_i u_{t-i} = \mu + \sum_{i=0}^{\infty} \Phi_i u_{t-i} \tag{3}$$

The challenge of estimating these parameters incurs the same difficulties as obtaining the parameters of the primitive model from the reduced model. This methodology does not permit estimation in cases of underidentification, where the number of equations is fewer than the number of unknowns. To identify the system written in MA form, we estimate each equation by OLS and then compute the Cholesky factorization of the reduced form VAR covariance matrix (Lütkepohl, 2005).

2.2. Model checking

2.2.1. Stationarity

The formal approach to ascertain the stationarity of time series involves employing the unit root test, with the Augmented Dickey-Fuller (ADF) test serving this purpose. The null hypothesis indicates the existence of a unit root, indicative of non-stationarity. Its rejection implies that the underlying stochastic process governing the series is time-invariant. In that case, the VAR model can be estimated with the series at level. Throughout the implementation of the ADF test, the analysis considered various model specifications, including those without a constant or trend, with a constant, and with both a constant and a trend.

2.2.2. Serial autocorrelation

To address the threat of serial autocorrelation, which has the potential to compromise the integrity of impulse response functions, the authors employ the Ljung and Box (1978) test. The number of lags, denoted as h, is determined by the Schwert (1989) criterion. In short, the null hypothesis posits that residuals up to the h-th lag exhibit white noise characteristics. When rejected, it implies that at least one autocorrelation is statistically different from zero. In such a case, the model must be rejected, prompting a re-specification and re-estimation of the VAR (Lütkepohl, 2011).

2.2.3. Model stability

To appraise the temporal consistency of a VAR model, it is imperative to scrutinize its stability, denoting the absence of explosions in the time series and the predictability of behavior over time. The stability examination aims to validate whether the roots of the eigenvalues in the coefficient matrix fall within the unit circle. If the eigenvalues modulus resides within the unit circle, the model attains stability, rendering the dependent variables weakly stationary. Conversely, if the eigenvalues modulus lies outside the unit circle, the model is deemed unstable, and the dependent variables are integrated.

According to Lütkepohl (2005), the stability of the VAR model assumes paramount significance in ensuring the reliability of results, as an unstable model can engender inaccurate predictions, particularly sensitive to minor fluctuations in input data. Furthermore, as highlighted by Baum (2013), adherence to stability conditions is imperative for the accurate estimation of Impulse Response Functions and Variance Decomposition.

2.3. Model evaluation

2.3.1. Impulse Response Function (IRF)

The IRF illustrates how a shock to a specific variable propagates to others over time, enabling the measurement of the magnitude and time horizon of the impact. In linear models with uncorrelated error terms, this process is straightforward. However, in the presence of correlated errors, identifying shocks with specific variables becomes complex and lacks clarity. This complexity arises from common components in errors affecting multiple variables. According to Farias (2008), a common approach in such cases is to arbitrarily assign the effects of these shared components to the variable that appears first in the system. The drawback of this approach is its dependency on the particular equation order within the model.

In a VAR(p) model, the derivation of IRFs involves inducing a one-period shock to an endogenous variable. Specifically, an increase in u_1 by one standard deviation at time $t = 0$ is considered, maintained for only one period, constituting a “boost". The impact of this shock permeates through the model, influencing all endogenous variables. Subsequently, a one-period shock is introduced to the next endogenous variable, for instance, an increase in u_2 by one standard deviation. This iterative process extends to all endogenous variables, allowing for a comprehensive tracking of effects across the entire model.

2.3.2. Forecast error variance decomposition (FEVD)

A complementary approach to analyzing the results of the VAR model is through variance decomposition. This technique enables the breakdown of forecast error variances for each variable into components attributable to the variable itself and others. The outcome is a percentage-wise representation of the impact that a shock to a specific variable exerts on itself and other system variables.

According to Enders (2008), if the shocks observed in a variable, y_2t, cannot account for the variance in the forecast error of variable y_1t, the sequence y_1t is deemed exogenous; otherwise, it is considered endogenous. Lütkepohl (2005) further notes that this technique is sensitive to changes in the considered system. The forecast error variance components may undergo alterations with system expansions, including additional variables or exclusions of series from the model. Furthermore, factors such as measurement errors, seasonal adjustments, and the use of aggregated data can impact the outcomes of this approach.

3. Data

This study utilizes monthly data from August 2010 to January 2023 (150 observations). The beginning of this time frame stems from the earliest available data on Bitcoin trading prices. The variables used in this study (in stationary form) are the following: returns on S&P500 (Ret_SP500), Gold (Ret_Gold), and bitcoin (Ret_BTC); first differences of the US 1y interest rates (Dif_US_1Y) and the VIX (Dif_VIX); and surprise inflation (Surp_Inf). Table 1 describes and details the variables considered in this study.

Our framework for variable selection, estimation, and inference is similar to Choi and Shin (2021), but we introduce modifications to enhance the identification of inflation shocks. These adjustments are inspired by the work of Kuttner (2001), Romer and Romer (2004), and Nakamura and Steinsson (2017), and the specification proposed by Lowenkron and Garcia (2007). Particularly, as described in Table 1, we include in the VAR system inflation surprises estimated as the difference between the actual (CPI_t) and the expected ($E_{t-1}(CPI_t)$) CPI MoM (also CPI YoY and Core PCE MoM in further analyses). With such refinements, we seek more assertive estimates and robust outcomes by identifying inflation through its surprising component.

One key operational concern is to homogenize each variable’s measurement period to coincide with inflation announcements, in the same spirit as Kuttner (2001), Cochrane and Piazzesi (2002), and Hanson and Stein (2015).[3] Thus, we consistently use the closing quote on the day of the inflation announcement relative to the closing price of the previous day for all series. By doing that, our variables capture the impact of the inflation announcement, regardless of different trading hours across various financial markets.

Particularly, the specification of variables, excluding *Surp_Inf*, involves calculating changes from $t - 1$ to t for each announcement of the U.S. economy price index, CPI or PCE (day t). This approach aims to capture the complete market response to the announcement, operating under the implicit assumption that the entire reaction to the economy price index announcement may not be immediate. Investor uncertainty about the implications of news may lead to a gradual reaction, where beliefs are updated as others’ interpretations unfold through trading volume, the pricing process, and financial media. Consistent with this idea and similar to our case, Hanson & Stein (2015) argues the full reaction to an FOMC announcement might not be instantaneous.[4]

Regarding the scale transformation applied to specific variables, this is a practice frequently adopted in this type of study. As Morettin and de Castro Toloi (2006) emphasize, working with scale-free returns rather than asset prices is recommended since the former have more advantageous statistical properties.

3.1. Descriptive analysis

As shown in Table 2, the centrality measures across all series are similar and nearly zero. Regarding asymmetry values, the return series exhibit negative asymmetry, with *Ret_Gold* approaching a distribution closest to symmetry. Kurtosis results depart from normal

[3] The authors use a window of one or two days around FOMC meetings to identify the effects of changes in monetary policy on financial markets.

[4] More specifically, Fleming and Remolona (1999) explored price dynamics in the financial market following the release of crucial economic announcements, such as monetary policy decisions. Their findings indicate a gradual formation of prices in the market after such information is released, characterized by elevated levels of volume and volatility persisting for at least 90 min.

Table 1
Data sources and description of model variables.

Variable	Calculation	Definition and source
Ret_SP500	$Ln(SP500_t/SP500_{t-1})$	The log-return is derived from a theoretical portfolio comprising the 500 largest companies traded on the NYSE or NASDAQ exchanges. It is widely used to capture general financial market conditions. Source: Yahoo Finance.
Ret_Gold	$Ln(Gold_t/Gold_{t-1})$	The log-return is obtained from gold futures contracts traded on the COMEX, a division of NYMEX. It serves as a crucial benchmark for investors seeking to track the price dynamics of gold in the financial market. Source: Yahoo Finance.
Ret_BTC	$Ln(BTC_t/BTC_{t-1})$	The log-return is calculated from the real-time market price of Bitcoin, determined by global transactions. Source: Investing.com.
Dif_US_1Y	$US1Y_t - US1Y_{t-1}$	Represents the change in the par yield of U.S. Treasury securities with a one-year maturity. It is based on the closing bid prices of the most recently auctioned securities in the over-the-counter market. The rate is published daily and reflects market expectations concerning the economic trajectory and monetary policy. Source: United States Department of the Treasury.
Dif_VIX	$VIX_t - VIX_{t-1}$	Represents the change in the reference indicator measuring the expected short-term volatility of the stock market. It is based on the stock options prices comprising the S&P500 index and is updated in real-time, during trading sessions. Source: Yahoo Finance.
Surp_Inf	$CPI_t - E_{t-1}(CPI_t)$	Represents the unexpected component of inflation, calculated at the moment of the official inflation rate announcement as the difference between the actual (CPI_t) and the expected ($E_{t-1}(CPI_t)$) CPI MoM. Alternative metrics such as CPI YoY and Core PCE MoM will also be taken into account. MoM (YoY) data are (are not) seasonally adjusted. Source: U.S. Bureau of Labor Statistics, Bureau of Economic Analysis and Investing.com.

Note: The subscript t ($t-1$) denotes, except for *Surp_Inf*, the closing price of the asset/index on the day (previous day) of the official inflation rate announcement.

Table 2
Summary statistics.

Variable	Mean	p50	SD	Asymmetry	Kurtosis	Min	Max	Obs.
Ret_SP500	−0.000	0.001	0.012	−0.680	6.759	−0.050	0.054	150
Dif_VIX	−0.123	−0.175	1.899	1.794	9.214	−5.710	11.090	150
Surp_Inf	−0.000	0.000	0.136	0.769	2.588	−0.400	0.600	150
Dif_US_1Y	0.003	0.000	0.045	2.607	13.714	−0.160	0.230	150
Ret_Gold	0.001	0.001	0.009	−0.210	1.341	−0.029	0.028	150
Ret_BTC	−0.004	0.000	0.091	−5.094	49.918	−0.849	0.288	150

Note: This table provides statistics for all series covering the period from 2010:08–2023:01. The Ret (Dif) prefix indicates that the series was derived from returns (differences) calculated within one-day windows around monthly CPI MoM announcements.

standards, with most demonstrating a leptokurtic distribution. *Surp_Inf* is the closest to a normal pattern, with a kurtosis of 2.59. Regarding volatility, it is noteworthy that the standard deviation of Bitcoin returns significantly exceeds that of gold, consistent with findings by Choi and Shin (2021), Smales (2022) and Phochanachan et al. (2022).

Since the surprise measures of inflation are key to our analysis, we take a closer look at them. Fig. A1, available in Appendix A, illustrates the frequency distribution graphs for the *Surp_Inf*, *Surp_Inf_YoY*, and *Surp_Inf_PCE* series. For the first two, the actual value aligned with market expectations approximately 33 % of the time. This proportion is notably lower compared to the *Surp_Inf_PCE* series, where 55 % of the time the inflationary surprise equaled zero. This discrepancy may be attributed to differing criteria, such as composition and disclosure dates (the PCE announcement occurs after the CPI release for the same period of reference; thus, market agents update beliefs following any surprise in CPI). We can see in the figures that the dispersion of surprises is larger for CPI MoM and CPI YoY than for PCE MoM. Furthermore, surprises in CPI are positively skewed – i.e., extreme values are more common in the right tail (positive ones).

Another important issue to discuss is the timing and correlation of these surprises. Fig. A2 shows the actual, forecast, and surprise (actual - forecast) values for our baseline measure of inflation (CPI MoM, Panel A) and for the further analyses (CPI YoY, Panel B, and PCE MoM, Panel C). We can observe from the Figure that average surprises are close to zero; however, positive surprises started to become more frequent in the recent period, starting in the middle of 2021, following a supply-side disruption brought by COVID-19.[5] Such a pattern is more pronounced in the CPI MoM and CPI YoY, suggesting that positive surprises in CPI are incorporated in PCE forecasts, reducing the forecast error in the latter inflation index.

Finally, Fig. A3 plots the three surprise measures and shows the correlation coefficient among them: the contemporaneous linear association is stronger for the pair CPI MoM-CPI YoY (0.63), but almost uncorrelated on the pairs CPI MoM-PCE MoM (0.12) and CPI YoY-PCE MoM (0.02). Such evidence reinforces that markets adapt expectations and reduce the forecast error for the PCE, whose announcement date always occurs later than the CPI. Finally, Fig. A4 shows the cross-correlation among each surprise pair and again reveals a clear pattern for CPI MoM-CPI YoY: the correlation coefficient is a lot stronger in t0 (i.e., their contemporaneous values). On

[5] On May 12, 2021 (reporting inflation from April 2021), the actual CPI MoM came on 0.8 %, while the expected inflation was 0.2 %. Such a surprise of 0.6 p.p. is the largest value in our sample period. After that, market agents witnessed a sequence of positive unexpected inflation values that made the FOMC start, on March 16, 2022, eight successive increases in the policy rate.

the other hand, the relationship between CPI and PCE is noisier, showing that the correlation is near zero for contemporaneous values and for other lead-lag relationships.

3.2. Model identification

Since the literature does not provide clear theoretical or empirical evidence on the causal relationships among all variables, we identify the structural shocks (i.e., the uncorrelated or orthogonal errors, not the reduced-form errors that correlate across equations) based on an economically reasonable temporal ordering criterion. This procedure represents a recursive VAR, where the error term in each regression equation is constructed to be uncorrelated with the error the preceding equations (Stock & Watson, 2001). Our ordering criteria for the Choleski triangular decomposition of the residuals are the following. Initially, variables representing broad measures of economic activity and market volatility are selected for their ability to assimilate new information promptly. Subsequently, series reflecting the dynamics of prices in the economy and the stance of monetary policy are incorporated. Lastly, variables theoretically presumed to be influenced by the aforementioned factors are considered: the price of a reference financial asset and the price of an emerging financial asset.

Given the relatively lower significance of Bitcoin in financial markets (Choi & Shin, 2021), as well as empirical findings suggesting its lesser efficiency compared to stock and gold markets in assimilating available information (Al-Yahyaee et al., 2018), which also indicate its responsiveness to changes in inflation, interest rates (Li & Wang, 2017), and financial market volatility (Bouri et al., 2017a), it is reasonable to treat Bitcoin as the most endogenous variable in the system.

Regarding the order of remaining variables, support for the identification assumption is drawn from the literature on the monetary VAR model, particularly highlighted by Choi and Shin (2021). Studies such as those by Christiano et al. (2005) and Coibion (2012) placed inflation before short-term interest rates in the Cholesky identification method applied in their analyses. Additionally, empirical research conducted by Beaudry and Portier (2006) and Jordà et al. (2017) provides evidence indicating that movements in financial variables, such as stock prices, may precede changes in macroeconomic variables like inflation and interest rates. The analyses by Rigobon and Sack (2003) and Kurov et al. (2022) are particularly relevant in this context, revealing the FED's responsiveness to the stock market dynamics, adjusting its monetary policy in response to changes in market conditions.

In light of the evidence provided, the baseline VAR model includes an intercept and six variables arranged in the following sequence: the log-return of the S&P500 (*Ret_SP*500), the variation of the VIX index (*Dif_VIX*), the surprise inflation (*Surp_Inf*), the variation in the one-year US Treasury bill yield (*Dif_US_*1*Y*), the log-return of gold prices (*Ret_Gold*), and the log-return of Bitcoin prices (*Ret_BTC*).

3.3. Model checking

The model-checking outcomes, featured in Appendix B, demonstrate favorable statistical properties, enhancing the reliability of the empirical results. Particularly, Table B1 exhibits the ADF test findings, indicating that all series have no unit root and are stationary at a 0.01 significance level. Such a result is robust to several model specifications – without drift or trend, with drift, with drift and trend, and with drift and linear and quadratic trends.

Furthermore, Table B2 displays the outcomes of the portmanteau test, revealing that, at a 0.05 significance level, only the VAR(6) model residuals exhibit white noise characteristics up to the maximum lag.[6] Finally, Table B3 showcases the findings of the stability diagnosis test applied to the VAR(6) model, indicating that all eigenvalues reside inside the unit circle, affirming the dynamic stability of the model.

4. Empirical results

4.1. Baseline model

Fig. 1 depicts the main empirical results of this study, considering the CPI MoM as the underlying index to calculate the inflation surprises. Specifically, the Figure shows the cumulative responses of each variable in the system following an impulse in the surprise component of the CPI MoM announcements.[7] Overall, we can infer that an unexpected, positive shock in CPI MoM increases Gold, BTC, and US1y interest rates, while it decreases the S&P 500 and has no effect on the VIX. While the results are statistically non-significant in most cases (the exception is the impact on interest rates in some horizons), such a pattern reveals that responses in BTC are similar to the ones of Gold and interest rates and contrast those of the S&P500.[8] Put differently, interest rates, Gold, and BTC returns increase following inflation shocks, while S&P returns diminish. Thus, the results suggest that Bitcoin could be a helpful hedge against unexpected fluctuations in inflation, an overall finding that aligns with empirical evidence supporting Bitcoin's role as an inflation hedge (Blau et al., 2021; Choi & Shin, 2021).

The baseline, full period results highlight further similarities in the behavior of BTC and Gold. Firstly, both assets exhibit an initial

[6] Defined by the Schwert (1989) criterion.

[7] Fig. C1 in Appendix C plots the entire set of impulse response functions for a comprehensive picture.

[8] Relative to BTC, The COIRFs suggest a similar, albeit less pronounced, response in gold returns to inflation shocks, indicating that the returns of both assets react in the same direction but with different intensities.

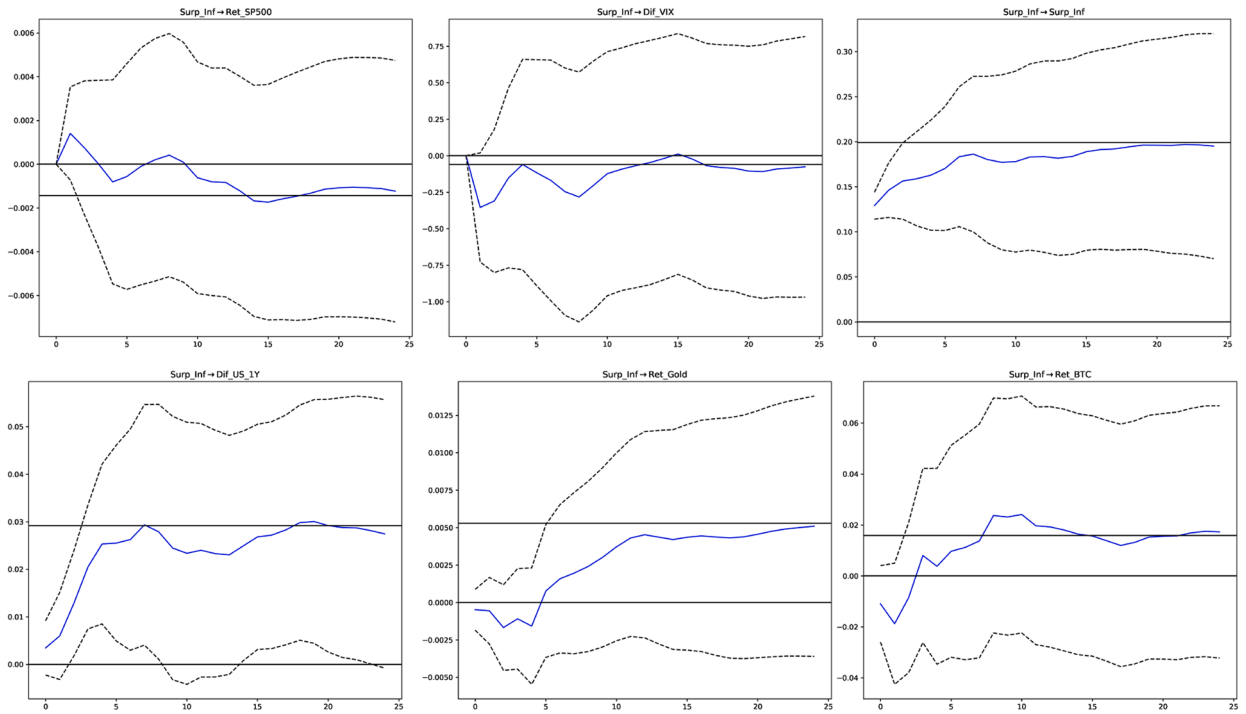


Fig. 1. COIRF - Baseline model (CPI MoM). Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of the six variables to the one-standard-deviation shock in inflation and their 95 % confidence bands for the sample period between 2010:08 and 2023:01. The Ret (Dif) prefix indicates that the series was constructed from returns (differences) calculated in one-day windows around monthly CPI MoM announcements. The horizontal axes show the number of months after the impulse.

negative effect, with Bitcoin displaying a more significant but less enduring impact. For both assets, there is a convergence to the final cumulative effect after the tenth month. Second, the forecast error variance decomposition of Bitcoin and gold returns (as shown in Fig. 2) indicates that Bitcoin (gold) return fluctuations are predominantly explained by a shock to itself, decreasing over time from 94 % (92 %) in the initial period to 69 % (64 %) in the final period. These results partially contrast with the findings of Choi and Shin (2021), who found this property only in gold. Regarding the contribution of a shock in inflation to the variance in the forecast error of the two series, gold is superior (8 %) compared to Bitcoin (6 %). This finding suggests that the relationship between return and inflation may be slightly more substantial for the commodity than for the cryptocurrency, aligning with the results of Smales (2022) and consistent with studies by Corbet et al. (2020) and Pyo and Lee (2020), which found that CPI announcements have a negligible impact on Bitcoin prices.

4.1.1. CPI MoM vs. CPI YoY

So far, we have considered inflation measured on a month-over-month (MoM) basis. One drawback of this approach is that short-term measures of inflation may be noisy. Thus, in this section, we replace the MoM CPI with the 12-month, year-over-year (YoY) CPI. This further analysis brings forth several potential benefits. Firstly, employing CPI YoY enables the capture of a more comprehensive perspective on price variations over time, providing a more representative view across an extended period. Additionally, it facilitates the identification of long-term inflation trends, as monthly fluctuations may be influenced by temporary factors that could distort the analysis. Hence, obtaining a more consistent view of inflationary pressures is possible by opting for an annual inflation measurement.

Fig. 3 illustrates that the results from cumulative orthogonalized impulse-response functions (COIRFs) are preserved even with the alternative specification of inflation shocks. Relative to the previous set of results (Fig. 1), the cumulative responses are very similar in terms of magnitude for both BTC and GOLD, reinforcing that they behave similarly.

4.1.2. CPI MoM vs. Core PCE MoM

Complementing the CPI, another relevant consumer price index in the U.S. is the Personal Consumer Expenditures (PCE), released by the Bureau of Economic Analysis (BEA). Although similar in spirit, the CPI and the PCE carry distinctions in four dimensions (Bureau of Labor Statistics (BLS), 2011): formula effect (Laspeyres vs. Fisher-Ideal), weight effect (Consumer Expenditure Survey vs. business surveys), scope effects (CPI considers all urban households, while the PCE considers goods and services consumed by all households, and nonprofit institutions serving households), and other effects (e.g., seasonal-adjustment differences, price differences, etc.). Furthermore, the PCE has been the official inflation target of the FOMC since January 2012, and looking at core inflation may eliminate temporary distortions as it excludes the volatile components of food and energy prices.

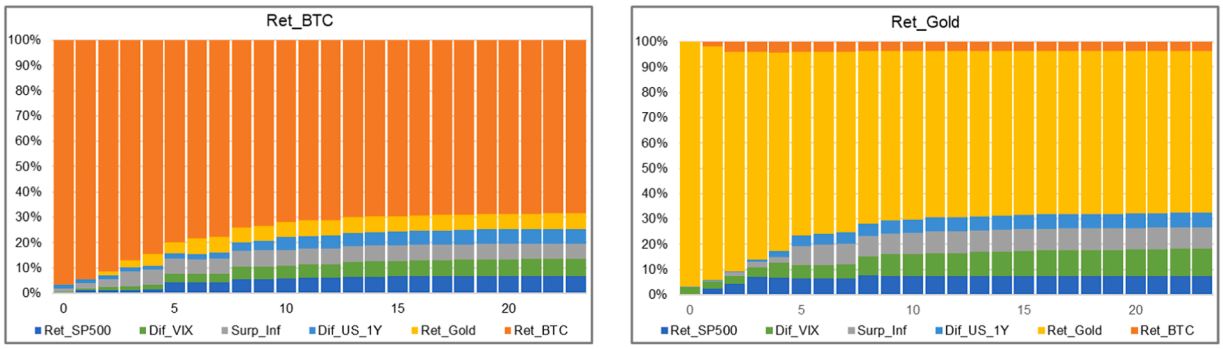


Fig. 2. FEVD - Baseline model. Note: This figure shows forecast error variance decomposition (FEVD) of Bitcoin and gold for the sample period between 2010:08 and 2023:01. The Ret (Dif) prefix indicates that the series was constructed from returns (differences) calculated in one-day windows around monthly CPI MoM announcements. The horizontal axes show the number of months after the impulse.

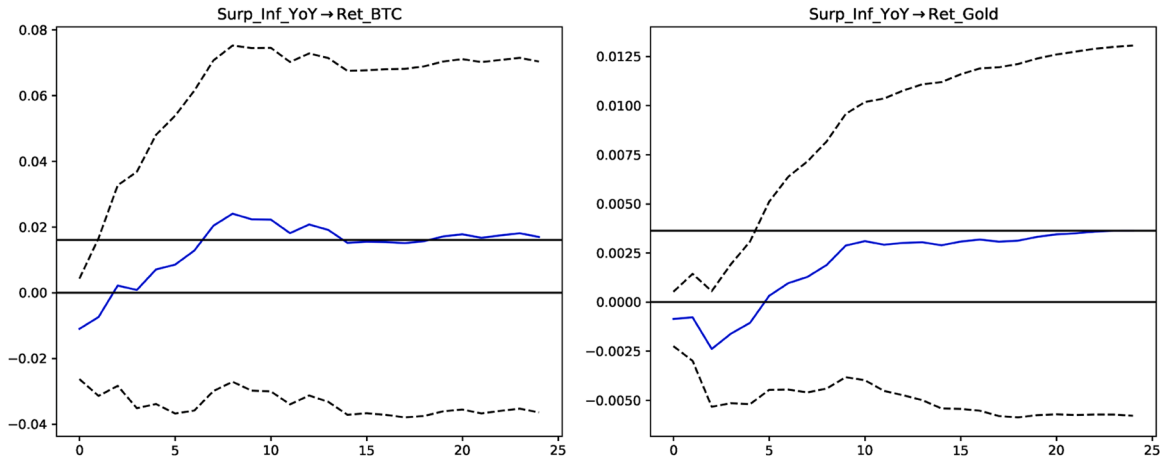


Fig. 3. COIRF - CPI YoY. Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of Bitcoin and gold to the one-standard-deviation shock in inflation and their 95 % confidence bands for the sample period between 2010:08 and 2023:01. The Ret prefix indicates that the series was constructed from returns calculated in one-day windows around monthly CPI YoY announcements. The horizontal axes show the number of months after the impulse.

Fig. 4 indicates that responses to inflation shocks measured by the PCE MoM generate partially contrasting results. While there is a shift in the response sign, BTC and Gold continue to react in the same direction. On the left graph, Bitcoin returns are observed to react with the same intensity, but negatively after a positive inflation shock. In contrast, the right graph shows that gold returns also respond negatively, although insignificantly, to the inflation shock. Furthermore, it is noteworthy that convergence to the final cumulative effect continues to occur after the tenth period.

The discrepancy in the response sign may be attributed to differing criteria, such as composition and disclosure dates, that can impact inflation surprises. Since PCE announcements occur after CPI releases, every surprise in the CPI – positive or negative – is plausibly incorporated into PCE expectations. This could explain why, as previously discussed, there is a very low contemporaneous correlation between surprise inflation in the pairs CPI MoM-PCE MoM and CPI YoY-PCE MoM.[9] Although such adjustments may explain the negative responses in BTC and Gold returns, it is imperative to note that the empirical validation of this mechanism is beyond the scope of the present study. However, this observation is suggested as a potential avenue for future research to deepen its understanding.

4.2. Robustness checks

4.2.1. Alternative VAR lag order

Acknowledging that estimated impulse responses may exhibit bias due to an insufficient number of lags (Lütkepohl, 2005; Enders,

[9] As Fig. A3 shows, correlations between CPI MoM (CPI YoY) and PCE MoM surprises are 0.12 (0.12) in our sample. They are not just low, on average, but they also differ in time trends: while the CPI has shown a sequence of positive surprises in 2021 onwards (following supply-chain imbalances brought by COVID-19 and, in 2022, the Russian invasion of Ukraine), the PCE has not shown such a trend.

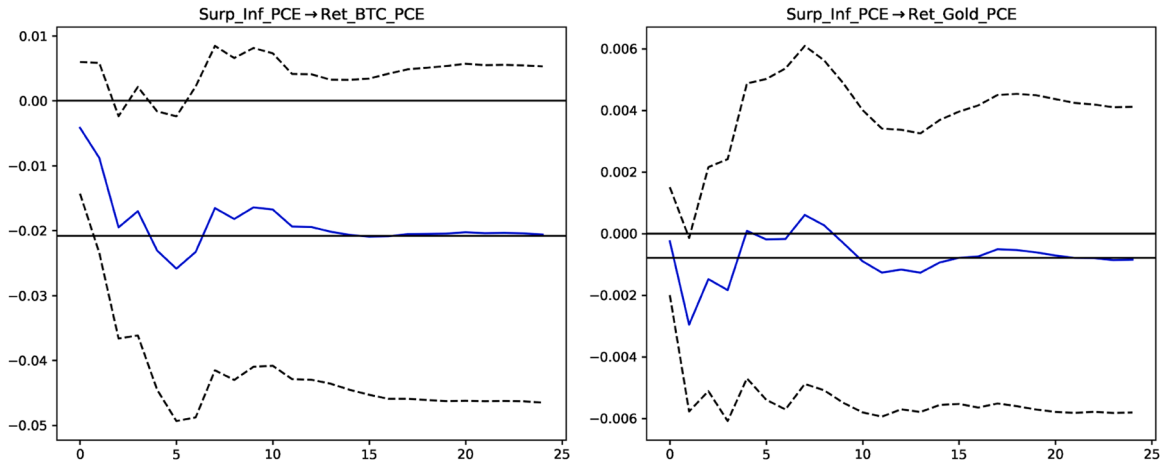


Fig. 4. COIRF - CPI MoM vs. Core PCE MoM. Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of Bitcoin and gold to the one-standard-deviation shock in inflation and their 95 % confidence bands for the sample period between 2010:08 and 2023:01. The Ret prefix indicates that the series was constructed from returns calculated in one-day windows around monthly Core PCE MoM announcements. The horizontal axes show the number of months after the impulse.

2008), we perform a sensitivity analysis by reestimating the model using nine and twelve lags. Fig. D1 in Appendix D illustrates the findings, highlighting that the general results remain consistent regardless of the lag length. However, we note that with extended lag, the impact of impulses on the results intensifies.

4.2.2. *Alternative order of variables in the Choleski decomposition*

The model is identified based on the premise that movements in the rest of the economy are less endogenous concerning the Bitcoin market. While this assumption is reasonable, it is crucial to note that any recursive assumption can pose challenges, especially in the presence of financial variables at a low frequency (Furlanetto et al., 2019). Furthermore, unless there are special reasons for a recursive structure, the order of the variables used in the Choleski decomposition of the white noise covariance matrix is arbitrary (Lütkepohl, 2005). Because there is no trivial solution to this issue, we re-estimate the model with modifications to the identification scheme.

Firstly, recognizing the inherent complexity of the relationship between inflation and interest rates and the possibility that this connection is bidirectional and variable in different economic contexts and market conditions (Alvarez et al., 2001), the time sequence of both series was inverted. Then, it was decided to position these variables at the beginning of the system, drawing on studies such as Roley & Sellon (1998), Bomfim (2003), Rigobon & Sack (2004) and Farka (2009) that illustrate the endogenous nature of interactions between the stock market, monetary policy, and inflation. Finally, Bitcoin was chosen to remain the most endogenous variable in the system, aligning with research by Bouri et al. (2017a), Li & Wang (2017), Al-Yahyaee et al. (2018), and Choi & Shin (2021). Therefore, the Cholesky ordering of the VAR system is as follows: *Dif_US_1Y*, *Surp_Inf*, *Ret_SP500*, *Dif_VIX*, *Ret_Gold*, and *Ret_BTC*.

Fig. D2, available in Appendix D, confirms that the change in model identification does not affect any of the baseline findings.

4.2.3. *Subsamples: analyses before and after COVID-19 and from 2014*

In this robustness check, we reestimate the baseline model using subsamples based on structural breaks identified for the bitcoin market. We briefly discuss the rationale for each of them before showing the results.

Pre-COVID-19 period. Although the COVID-19 pandemic is a unique opportunity to test Bitcoin's inflation-hedging properties, it is crucial to acknowledge potential influences from this atypical event.[10] As highlighted by Maneejuk et al. (2021), an asset's ability to hedge against inflation may be subject to the state of the economy, encompassing both stable and turbulent economic regimes. Furthermore, Smales (2022) identifies a structural break in BTC returns around the second week of March 2020. Thus, we re-estimate the model using data up to February 2020.

From 2014 onwards. Over the last decade, perceptions of cryptocurrencies, especially Bitcoin, have undergone significant transformations, impacting the trading system and market dynamics. In the initial phase of the examined period, Bitcoin trading volumes were minimal, and the cryptocurrency was not widely recognized as an investment option. Identifying the precise moment when Bitcoin emerged as a viable investment alternative is challenging. However, for this test, we follow the Choi and Shin (2021) proposition of a structural break in the Bitcoin market around 2013.

COVID-19 period. As Smales (2022) identifies a structural break in BTC returns between 9 March 2020 and 13 March 2020 with a high degree of confidence, we split the sample before and after this threshold. Thus, we also perform a robustness test using this subsample (From March 2020 to the end of the sample). One weakness of this analysis is the small sample period from COVID-19

[10] Previous research has shown that the sensitivity of performance measures to the COVID-19 period was higher for portfolios with cryptocurrency relative to otherwise identical portfolios (Colombo et al., 2021).

onwards, resulting in wider confidence intervals in the impulse-response analysis and requiring caution when analyzing the results.

Fig. D3 in Appendix D shows the contrasting results among different time periods. The cumulative response of Bitcoin was positive before COVID-19, decreased slightly from the Pre-COVID-19 to the 2014 onwards period, and became negative for the period from COVID-19 onwards. It is important to note that the sample size after COVID-19 is small, so any conclusions drawn should be made with caution. However, these results suggest that Bitcoin's ability to hedge against inflation vanished at a time when global inflation concerns and uncertainties became more apparent. On the other hand, Gold showed the opposite behavior: its cumulative responses were negative before COVID-19, turned positive from 2014 onwards, and increased during the COVID-19 period. In other words, the inflation-hedge property of gold increases when it is most needed.

Taken together, these findings align with empirical evidence from Phochanachan et al. (2022) and Smales (2022), suggesting that Bitcoin's reactions to inflation shocks are more significant in stable scenarios, unlike gold, which demonstrates better performance in turbulent contexts.

4.2.4. Benchmark assets other than Gold

So far, we have benchmarked bitcoin with gold, a well-known asset that protects investors against inflation. We have also analyzed equities since the S&P500 composes the system of equations. However, one should question whether other asset classes, like corporate bonds and real estate, may behave in a similar context. To answer that, we included the iShares Core U.S. Aggregate Bond ETF (AGG) and the Vanguard Real Estate ETF (VNQ) to proxy for corporate bonds and real estate, respectively. Following Choi and Shin (2021) and to keep our system of variables as parsimonious as possible, we have rerun the baseline model twice, replacing the benchmark asset from Gold to AGG and from Gold to VNQ, respectively.

In Fig. D4, the results show that, for the entire sample, the reactions of Corporate Bond returns are similar to those of the S&P500 and differ from those of BTC and Gold. Conversely, the reactions of Real Estate closely resemble those of Gold and BTC, suggesting it may serve as an inflation-hedging financial instrument. Expectedly, the reactions of BTC remain positive and display a similar pattern to the baseline model with Gold as the reference asset.

4.2.5. Including observed inflation in the model

Including a variable representing realized inflation enhances our understanding of the formation of expectations over time and their impact on economic behavior (Armantier et al., 2011; Weber et al., 2022; Binder & Kamdar, 2022), offering a more thorough analysis of the dynamic relationships between inflation and the price of assets like Bitcoin. Moreover, this approach provides a more comprehensive assessment of market conditions by considering the cumulative effects of inflation and structural changes in the economy.

Fig. D5, available in Appendix D, illustrates the key empirical findings of this examination. Firstly, actual inflation increases over time in response to a positive inflation shock. Secondly, Bitcoin returns experience a significant upswing following a positive shock to realized inflation, aligning with empirical evidence that supports Bitcoin's potential as an effective hedge against inflation (Blau et al., 2021; Choi & Shin, 2021). Lastly, gold returns display limited responsiveness to the realized inflation shock.

5. Concluding remarks

Cryptocurrencies have solidified their position as an asset class. Meanwhile, their alleged role as a hedge against inflation remains a question without a clear answer. We contribute to this debate by analyzing the dynamic response of bitcoin returns to inflation surprises, considering different inflation indices, changes in the sample period, and various model specifications. Following Smales (2022), but relying on a VAR framework for estimation and inference, we identify inflation shocks as *surprises*: the difference between the announced inflation and the analysts' consensus. By focusing on the unexpected aspect of CPI and Core PCE releases, our innovations consider all observed and unobserved factors that market agents take into account in their inflation projections, not just a few variables included in typical VAR systems.

The results suggest that Bitcoin can act as a hedge against unexpected fluctuations in inflation. However, the efficacy of Bitcoin as an inflation hedge appears contingent upon the selected price index (evidence is found on CPI MoM and CPI YoY, but not for the Core PCE MoM) and the analyzed sample period. Regarding the latter, the inflation-hedging properties of Bitcoin diminish when we exclude the initial sampling period ("early days" of Bitcoin) and vanishes when we consider just the subperiod following the outbreak of the COVID-19 pandemic onwards. Gold, on the other hand, presented better hedging properties after the outbreak of the COVID-19 pandemic, precisely when inflation concerns and uncertainties surged worldwide. One possible explanation for these contrasting time-varying performances is that mainstream adoption may be driving BTC returns to be closer to returns of other risky assets (like stocks), undermining its role as a hedge against inflation. At the end of the day, bitcoin does not seem to be as reliable as Gold in protecting portfolios against unexpected, positive inflation shocks.

Our paper expands the still incipient literature on the role of Bitcoin as an inflation hedge (see, e.g., Blau et al., 2021; Choi & Shin, 2021; Matkovskyy & Jalan, 2020; Conlon et al., 2021; Smales, 2022; Phochanachan et al., 2022) by analyzing the dynamic cumulative response of bitcoin returns to inflation surprises in a wide range of indices, methodological choices, and circumstances. Furthermore, this research carries direct implications for managers,[11] investors, and monetary authorities, contributing to the growing but still

[11] Not only asset managers, but also corporate managers since many corporations acquired bitcoin in recent years to manage excess cash reserves (Gimenes et al., 2023).

underexplored literature about the inflation-hedging properties of cryptocurrencies. While this study has unveiled numerous empirical findings, it is imperative to acknowledge certain caveats that offer valuable insights for future research. Firstly, the dynamic nature of the cryptocurrency market requires caution in interpreting results. Specifically, the employed model does not account for the potential impacts of regulatory changes or the proliferation of new cryptocurrencies on Bitcoin prices. Secondly, in comparison to prior analyses of other inflation-hedge assets, such as gold, the sampling period is confined to the cryptocurrency market's early stages. Consequently, pivotal aspects like trading volume or liquidity may undergo abrupt changes in the future, posing challenges to the drawn conclusions. Thirdly, the study's methodology relies on a linear model, potentially overlooking nonlinear relationships and structural shifts in the analyzed variables. Fourthly, our analysis does not consider the potential side effects of incorporating bitcoin into a diversified portfolio (for example, increased volatility and fat tails). Lastly, it is crucial to highlight that returns are calculated on a daily basis, which may lead to the loss of pertinent intraday information and impact analysis accuracy. Therefore, future studies should explore the feasibility of capturing these variations at a high frequency.

CRediT authorship contribution statement

Harold Rodriguez: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jefferson Colombo:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

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Appendix A. Descriptive statistics - Surprise inflation measures

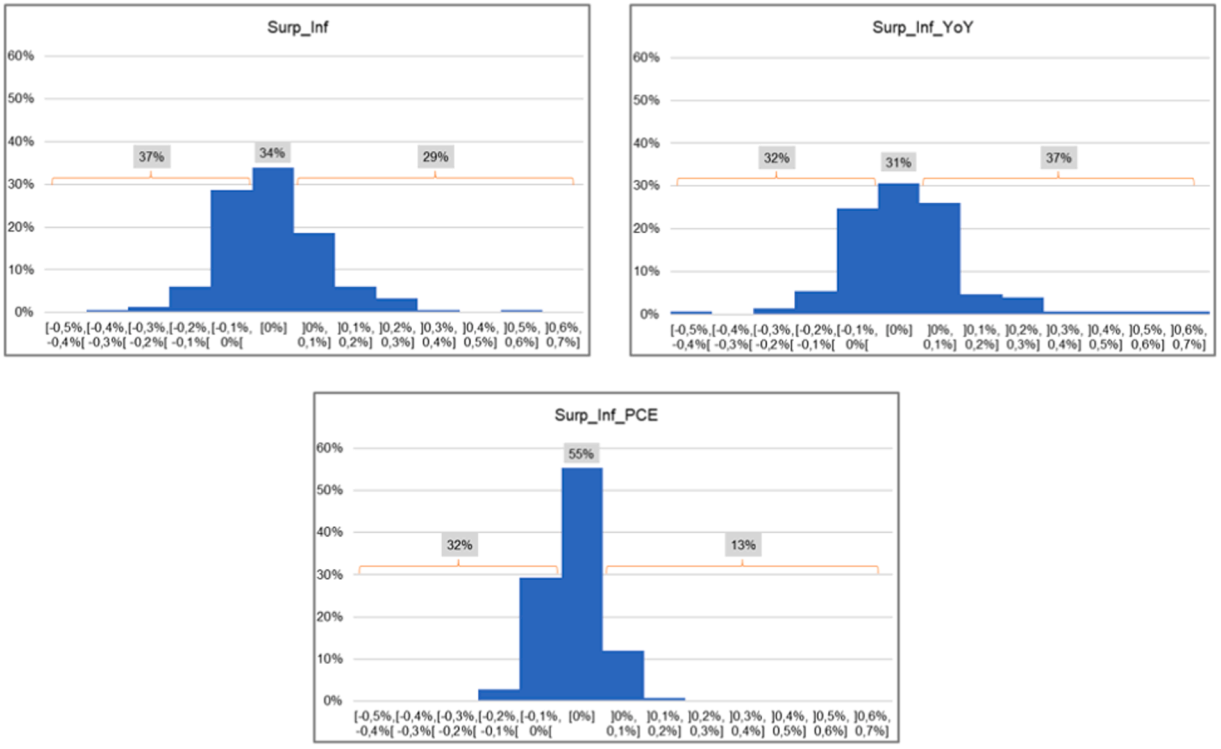
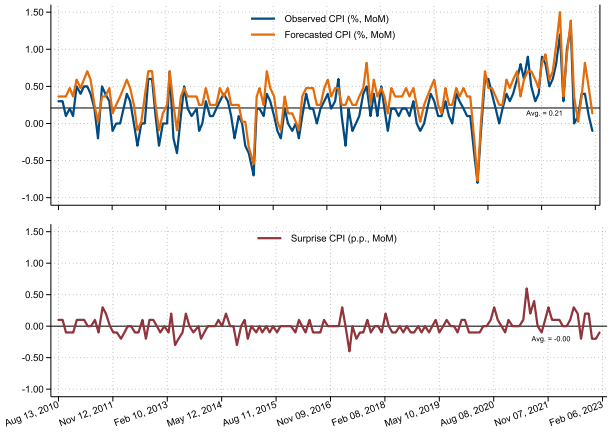
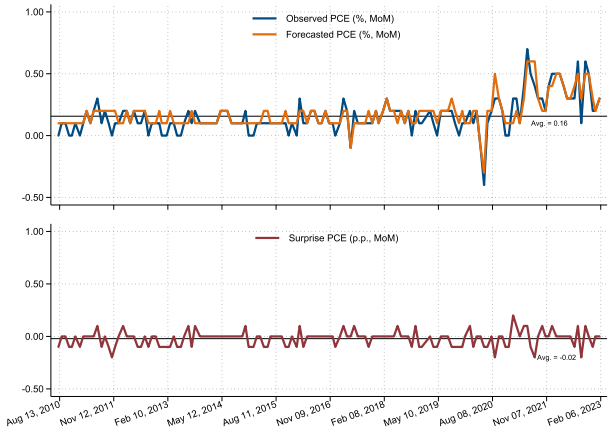


Fig. A1. Frequency distribution of surprise inflation. Note: This figure shows the frequency distribution graphs of the unexpected component of inflation calculated in one-day windows around monthly CPI MoM (top left graph), CPI YoY (top right graph), and Core PCE MoM (bottom graph) announcements. The sample period covers from 2010:08–2023:01. The horizontal axes represent different intervals/values of surprise inflation.

(a) CPI MoM – Actual, Forecast, and Surprise



(b) PCE MoM – Actual, Forecast, and Surprise



(c) CPI YoY – Actual, Forecast, and Surprise

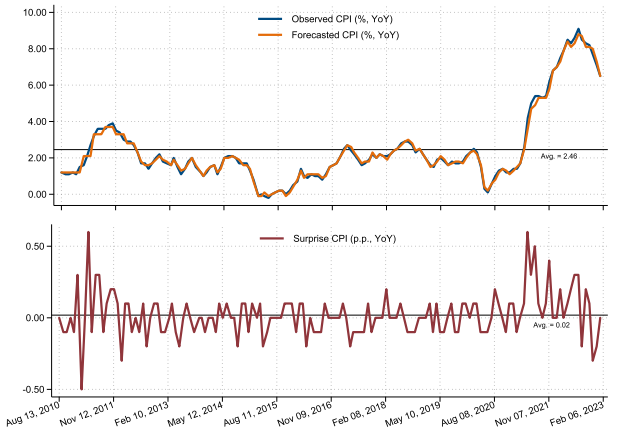


Fig. A2. CPI MoM, CPI YoY, and PCE MoM - Actual, Forecast, and Surprise comparisons. Note: Subfigures (a), (b), and (c) show the observed and the forecasted U.S. Consumer Price Index (CPI, %, Year-over-Year) for each release of the U.S. Bureau of Labor Statistics (BLS). Actual data is seasonally adjusted for MoM comparisons (not seasonally adjusted for YoY comparisons) and comes from the BLS, and forecasts are obtained at Investing.com. The lower part of the figure shows the time series of the Surprise component (difference between the observed, actual inflation, and the analysts' forecast). Covered CPI (PCE) announcements range from August 13, 2010 (August 30, 2010), to January 12, 2023 (Jan 27, 2023).

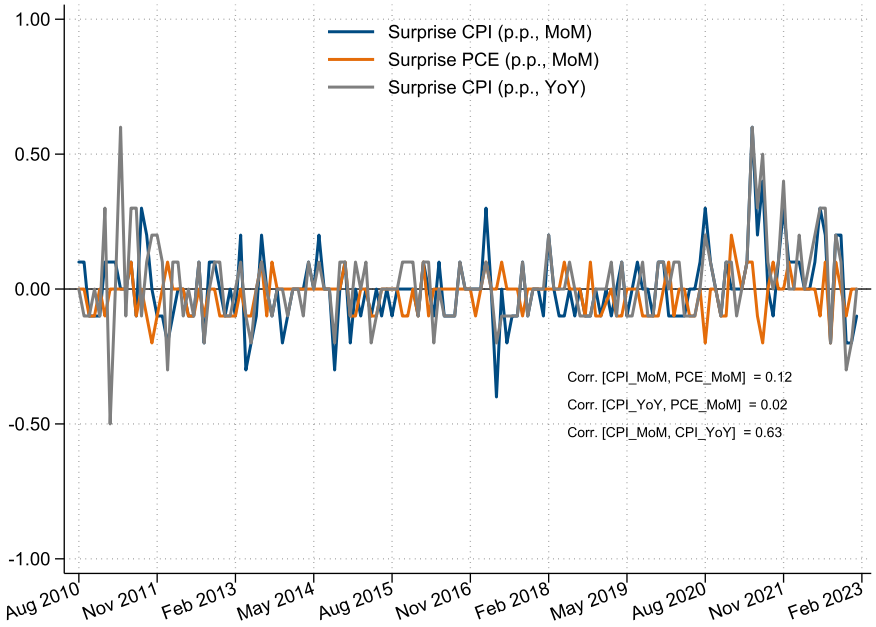


Fig. A3. Time series and correlation of Surprises – CPI MoM, CPI YoY, and PCE MoM. Note: The upper part of this figure shows the observed and forecasted U.S. Consumer Price Index (CPI, %, Year-over-Year) for each release of the U.S. Bureau of Labor Statistics (BLS). Actual data is seasonally adjusted for MoM comparisons (not seasonally adjusted for YoY comparisons) and comes from the BLS, and forecasts are obtained at Investing.com. The lower part of the figure shows the time series of the Surprise component (difference between the observed, actual inflation, and the analysts' forecast). Covered CPI (PCE) announcements range from August 13, 2010 (August 30, 2010), to January 12, 2023 (Jan 27, 2023).

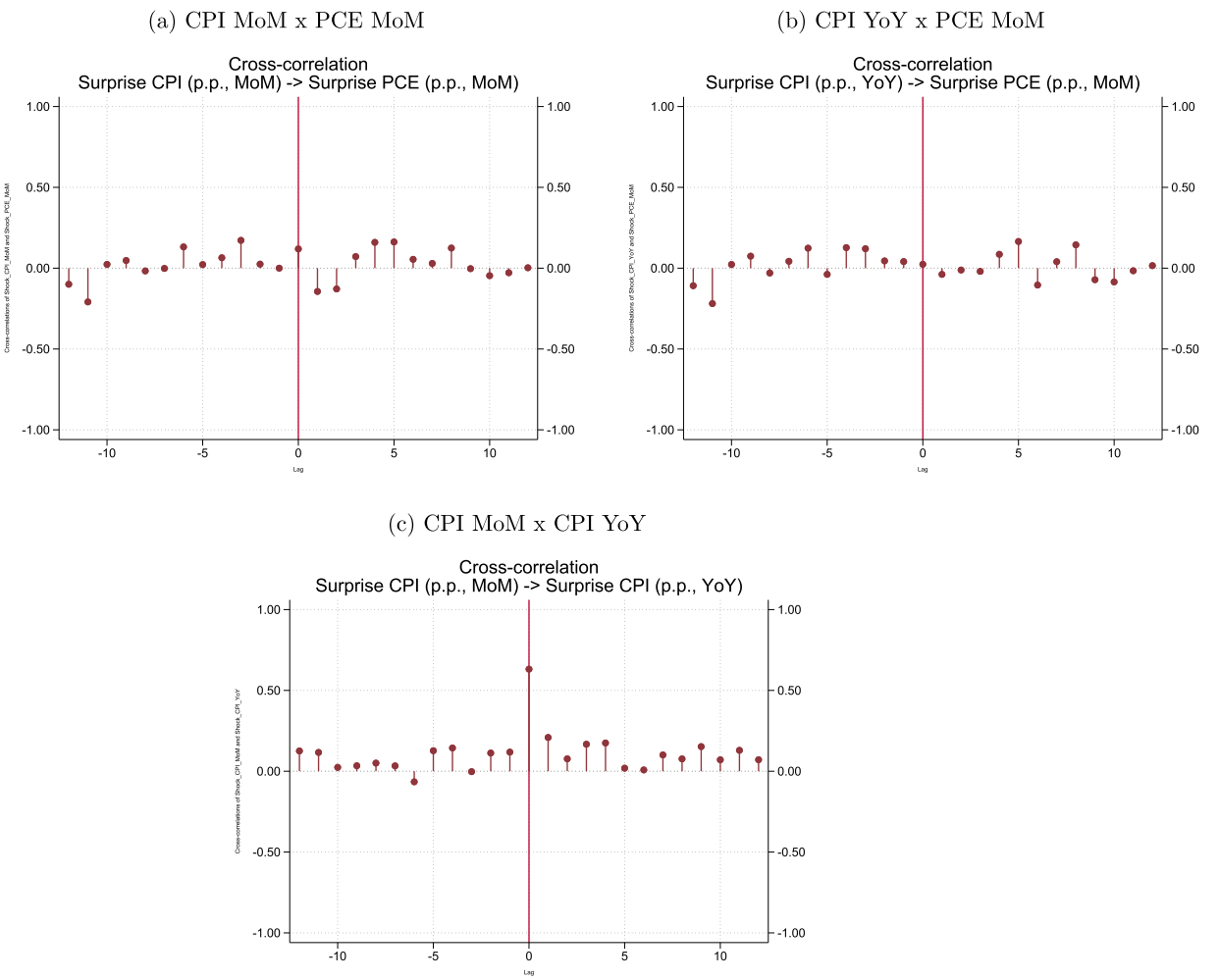


Fig. A4. Cross-correlation between surprise inflation measures: CPI MoM, PCE MoM, and CPI YoY. Note: This Figure shows the cross-correlation among each pair of computed inflation surprise (difference between the observed, actual inflation, and the analysts' forecast): CPI MoM-PCE MoM (Panel A), CPI YoY-PCE MoM (Panel B), and CPI MoM-CPI YoY (Panel C). The Y-axis shows the linear correlation coefficient that ranges from -1 to $+1$. The X-axis presents the lags and leads that are symmetrical and range from one to ten. The red vertical line emphasizes the contemporaneous linear correlation coefficient (lead/lag equals zero). Covered CPI (PCE) announcements range from August 13, 2010 (August 30, 2010), to January 12, 2023 (Jan 27, 2023).

B. Model Checking

Table B1
Augmented Dickey-Fuller unit root test.

Variable	Order	Lags	Test stat.	p-value	Assessment
			Panel A: Model with drift		
Ret_SP500	I(0)	4	−5.868 *	< 0.01	Reject H_0
Dif_VIX	I(0)	0	−13.547 *	< 0.01	Reject H_0
Surp_Inf	I(0)	0	−10.074 *	< 0.01	Reject H_0
Dif_US_1Y	I(0)	10	−4.508 *	< 0.01	Reject H_0
Ret_Gold	I(0)	0	−10.243 *	< 0.01	Reject H_0
Ret_BTC	I(0)	0	−11.729 *	< 0.01	Reject H_0
			Panel B: Model without drift or trend		
Ret_SP500	I(0)	4	−5.894 *	< 0.01	Reject H_0
Dif_VIX	I(0)	0	−13.527 *	< 0.01	Reject H_0
Surp_Inf	I(0)	0	−10.108 *	< 0.01	Reject H_0

(*continued on next page*)

Table B1 (*continued*)

Variable	Order	Lags	Test stat.	p-value	Assessment
Dif_US_1Y	I(0)	10	−4.488 *	< 0.01	Reject H_0
Ret_Gold	I(0)	0	−10.075 *	< 0.01	Reject H_0
Ret_BTC	I(0)	0	−11.745 *	< 0.01	Reject H_0
			Panel C: Model with drift and trend		
Ret_SP500	I(0)	4	−5.836 *	< 0.01	Reject H_0
Dif_VIX	I(0)	0	−13.554 *	< 0.01	Reject H_0
Surp_Inf	I(0)	0	−10.345 *	< 0.01	Reject H_0
Dif_US_1Y	I(0)	10	−4.872 *	< 0.01	Reject H_0
Ret_Gold	I(0)	0	−10.898 *	< 0.01	Reject H_0
Ret_BTC	I(0)	0	−11.688 *	< 0.01	Reject H_0
			Panel D: Model with drift and linear and quadratic trends		
Ret_SP500	I(0)	4	−5.895 *	< 0.01	Reject H_0
Dif_VIX	I(0)	0	−13.523 *	< 0.01	Reject H_0
Surp_Inf	I(0)	0	−10.669 *	< 0.01	Reject H_0
Dif_US_1Y	I(0)	10	−5.308 *	< 0.01	Reject H_0
Ret_Gold	I(0)	0	−10.876 *	< 0.01	Reject H_0
Ret_BTC	I(0)	0	−11.801 *	< 0.01	Reject H_0

Note: This table shows the results of the Augmented Dickey-Fuller (ADF) test for the Model with different specifications. The sample period covers from 2010:08–2023:01. The Ret (Dif) prefix indicates that the series was constructed from returns (differences) calculated in one-day windows around monthly CPI MoM announcements. The maximum number of lags considered for implementing the test is 13 and is defined by the Schwert criterion (1989). *, **, and *** denote statistical significance at 1, 5, and 10 %, respectively, and lead to the rejection of the null hypothesis indicating the non-existence of a unit root. The Lags column reports the optimal number of lags following the Akaike information criterion (AIC).

Table B2
Portmanteau residual autocorrelation test.

Model	Test stat.	Critical value (5 %)	p-value	Degrees of freedom	Assessment
VAR(6)	264.8	290.0	0.277	252	Accept H_0
VAR(7)	265.5	251.3	0.012	216	Reject H_0
VAR(8)	265.5	212.3	0.000	180	Reject H_0

Note: This table shows the results of the Ljung and Box test (1978) for the Model with different VAR orders. The sample period covers from 2010:08–2023:01. The maximum number of lags considered for implementing the test is 13 and is defined by the Schwert criterion (1989). Q_{LB} statistic values lower than the critical values lead to the acceptance of the null hypothesis and indicate that the residual autocorrelation up to the maximum number of lags is zero, considering a 95 % confidence interval.

Table B3
Model stability.

Lag	Ret_SP500	Dif_VIX	Surp_Inf	Dif_US_1Y	Ret_Gold	Ret_BTC
1	0.120	0.233	0.233	0.235	0.077	0.077
2	0.207	0.185	0.185	0.152	0.068	0.008
3	0.631	0.157	0.157	0.158	0.158	0.032
4	0.242	0.223	0.223	0.206	0.206	0.169
5	0.197	0.197	0.273	0.120	0.107	0.034
6	0.271	0.334	0.076	0.076	0.011	0.093

Note: This table shows the results of the stability diagnosis test applied to the VAR(6) model. The sample period covers from 2010:08–2023:01. The Ret (Dif) prefix indicates that the series was constructed from returns (differences) calculated in one-day windows around monthly CPI MoM announcements. The values reported correspond to the eigenvalues (in modulus) associated with each model lag. Values lower than 1 mean that they are inside the unit circle, which highlights the stability of the model.

C. Full results - COIRFs of the baseline model

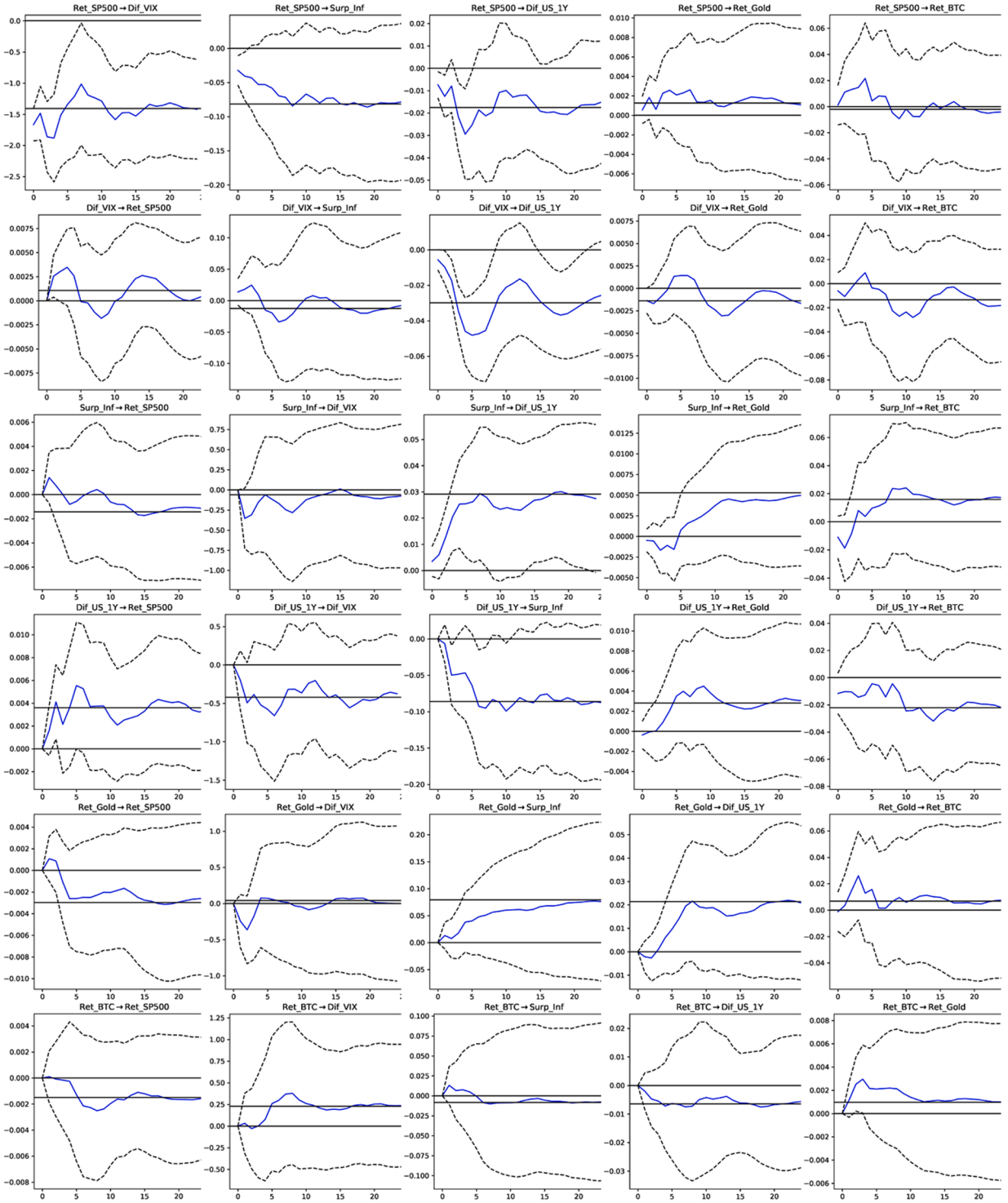


Fig. C1. COIRF Panel - Baseline model. Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of the six variables and their 95 % confidence bands for the sample period between 2010:08 and 2023:01. The Ret (Dif) prefix indicates that the series was constructed from returns (differences) calculated in one-day windows around monthly CPI MoM announcements. The units of the horizontal axes are months.

D. Robustness Tests

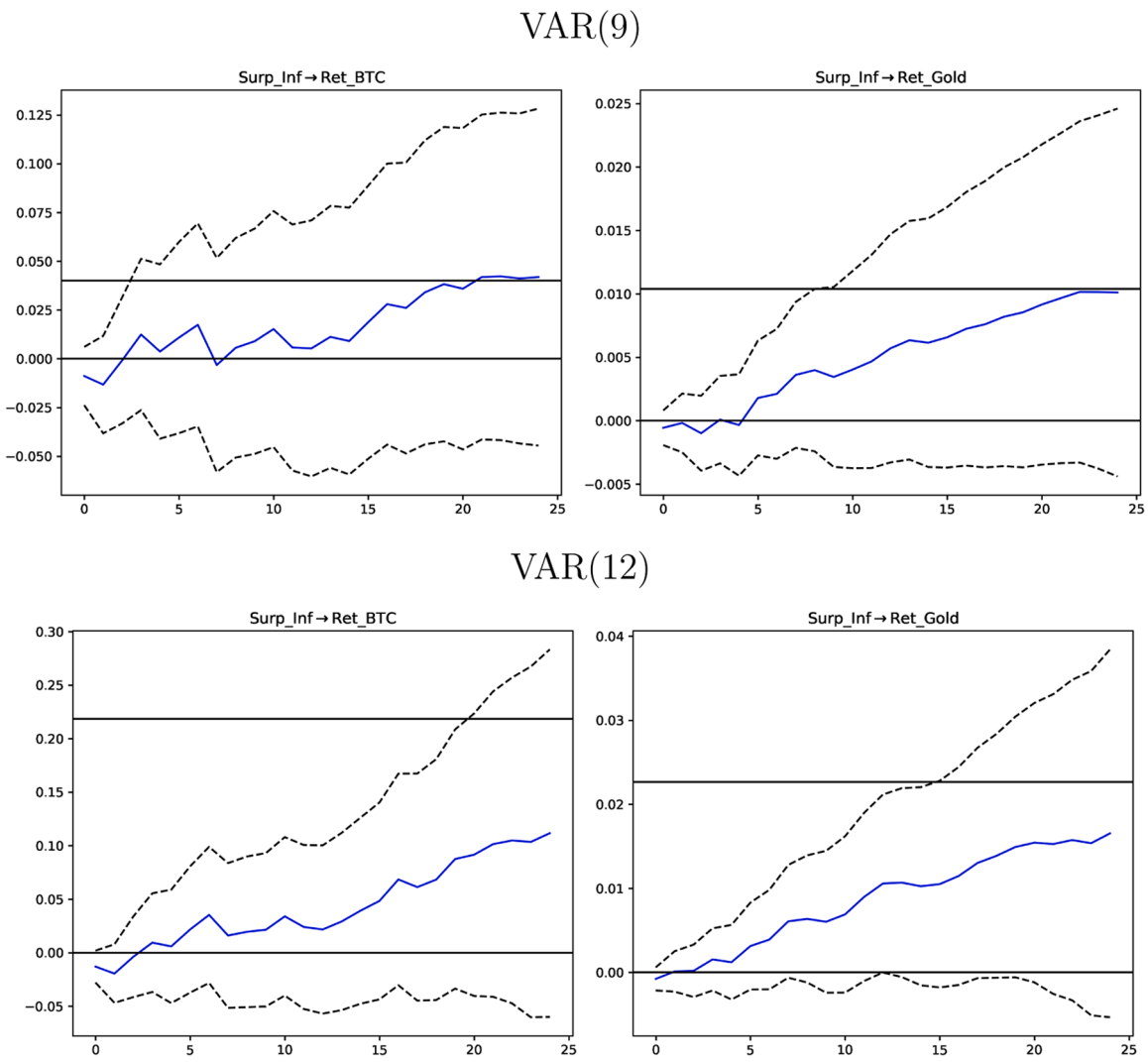


Fig. D1. RT 01 - Alternative VAR order. Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of Bitcoin and gold to the one-standard-deviation shock in inflation and their 95 % confidence bands for the sample period between 2010:08 and 2023:01 for the Model with alternative lags. The Ret prefix indicates that the series was constructed from returns calculated in one-day windows around monthly CPI MoM announcements. The horizontal axes show the number of months after the impulse.

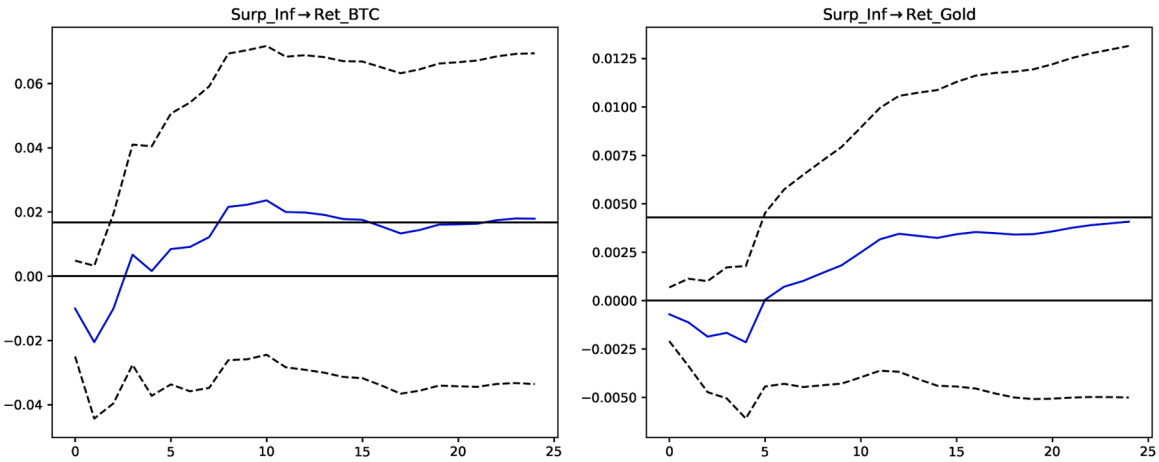


Fig. D2. RT 02 - Alternative order of variables in the Choleski decomposition. Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of Bitcoin and gold to the one-standard-deviation shock in inflation and their 95 % confidence bands for the sample period between 2010:08 and 2023:01 for the Model with the following temporal ordering: *Dif_US_1Y*, *Surp_Inf*, *Ret_SP500*, *Dif_VIX*, *Ret_Gold* and *Ret_BTC*. The Ret (Dif) prefix indicates that the series was constructed from returns (differences) calculated in one-day windows around monthly CPI MoM announcements. The horizontal axes show the number of months after the impulse.

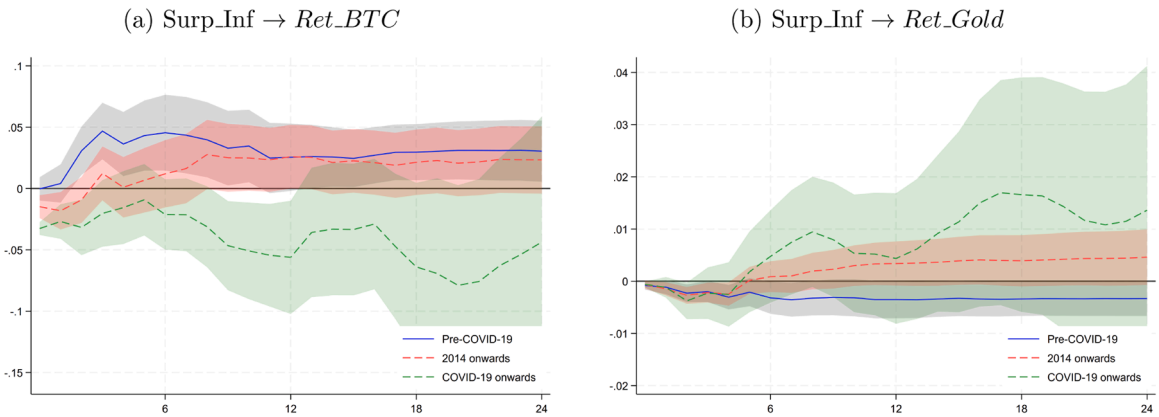


Fig. D3. Subperiods: Before COVID-19, COVID-19 onwards, and after the 2014 BTC structural break. Note: Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of Bitcoin and gold to the one-standard-deviation shock in inflation surprise and their 68 % confidence bands for three sample periods: Pre-COVID-19 (2010:08–2020:02), 2014 onwards (2014:01–2023:01), and COVID-19 onwards (2020:03–2023:01). The Ret prefix indicates that the series was constructed from returns calculated in one-day windows around monthly CPI MoM announcements. The horizontal axes show the number of months after the impulse.

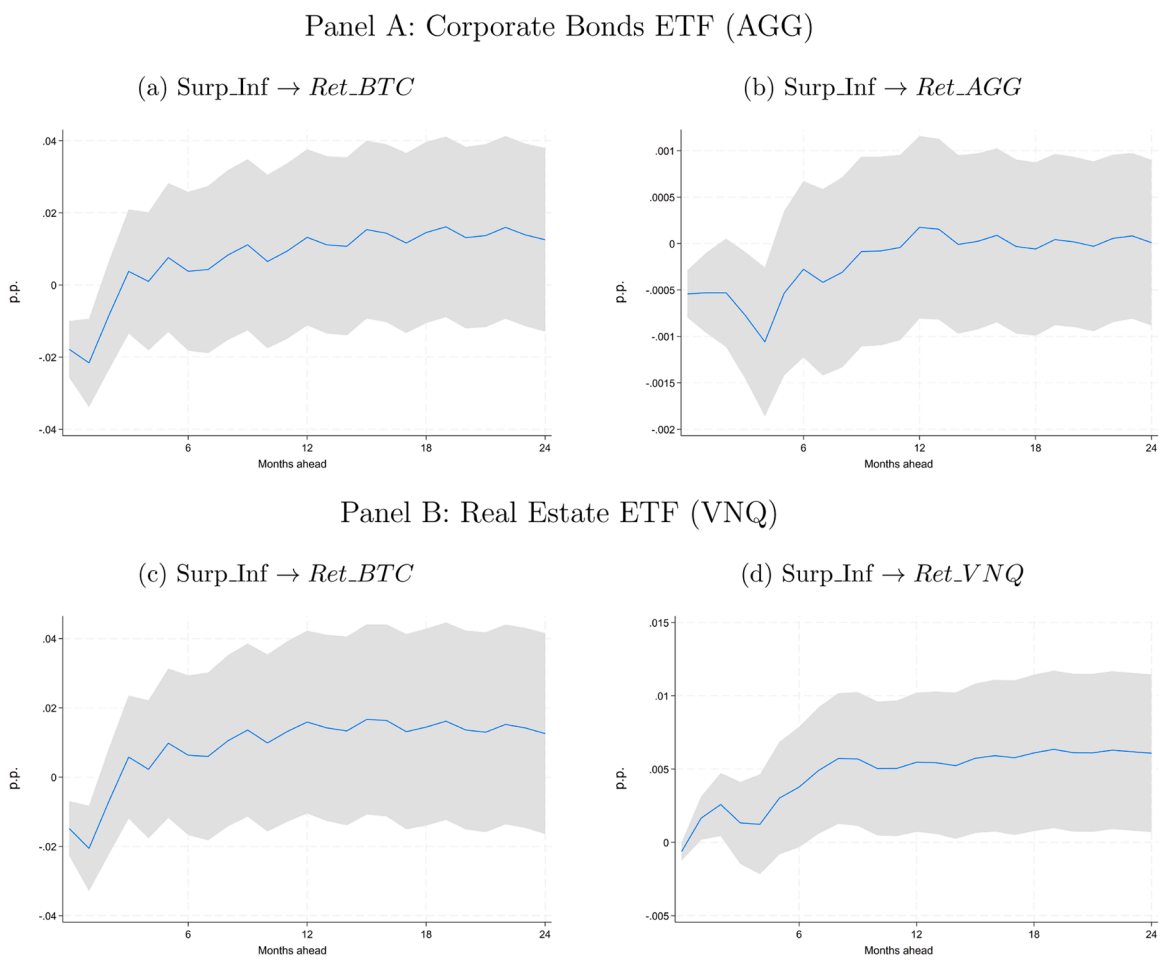


Fig. D4. Alternative benchmarks to gold: corporate bonds and real estate. Panel A: Corporate Bonds ETF (AGG). Panel B: Real Estate ETF (VNQ). Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) of bitcoin and Corporate Bonds (Panel A) and bitcoin and Real Estate (Panel B) to the one-standard-deviation shock in inflation and their 68 % confidence bands for the sample period between 2010:08 and 2023:01. The Ret prefix indicates that the series was constructed from returns calculated in one-day windows around monthly CPI MoM announcements. The units of the horizontal axes represent months.

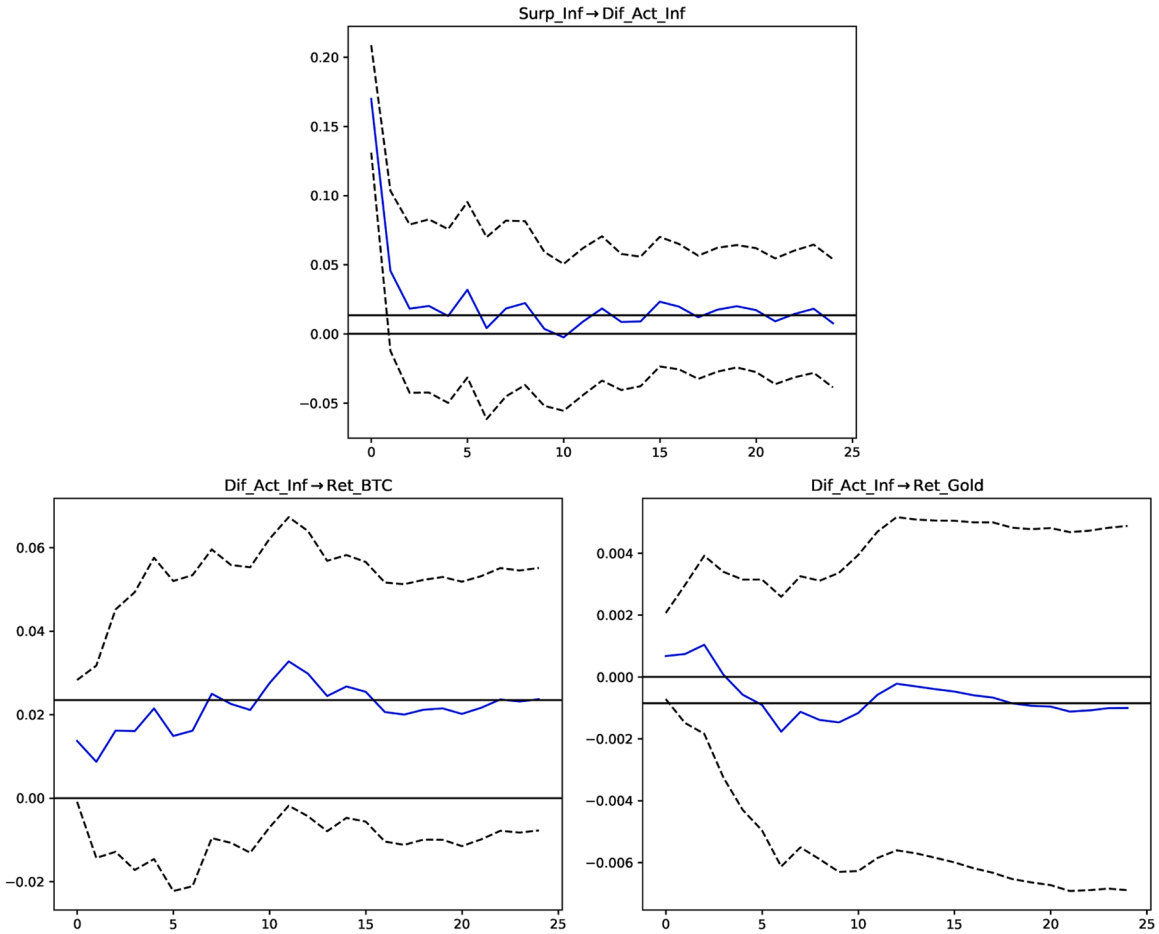


Fig. D5. RT 06 - Including observed inflation in the VAR system. Note: This figure shows cumulative orthogonal impulse response functions (COIRFs) and their 95 % confidence bands for the sample period between 2010:08 and 2023:01 for the Model with the inclusion of the actual inflation series (*Act_Inf*). The Ret (Dif) prefix indicates that the series was constructed from returns (differences) calculated in one-day windows around monthly CPI MoM announcements. The units of the vertical axes correspond to the intensity of the series' response to the one-standard-deviation shock in inflation (actual inflation). The units of the horizontal axes are a month.

Table D1
Summary statistics: Full period, Before COVID-19 and After BTC structural break.

Variable	Mean	p50	SD	Asymmetry	**Kurtosis**	Min	Max	Obs.
				Panel A: Full period (2010m8 to 2023m1)				
Ret_SP500	−0.000	0.001	0.012	−0.680	6.758	−0.050	0.054	150
Dif_VIX	−0.123	−0.175	1.900	1.794	9.214	−5.710	11.090	150
Surp_Inf	−0.000	0.000	0.136	0.769	2.588	−0.400	0.600	150
Dif_US_1Y	0.003	0.000	0.045	2.607	13.714	−0.160	0.230	150
Ret_Gold	0.001	0.001	0.009	−0.210	1.341	−0.029	0.028	150
Ret_BTC	−0.004	0.000	0.091	−5.094	49.918	−0.849	0.288	150
				Panel B: Before COVID−19 (2010m8 to 2020m1)				
Ret_SP500	0.001	0.001	0.009	−1.467	6.692	−0.046	0.020	114
Dif_VIX	−0.167	−0.140	1.703	2.134	17.015	−5.710	11.090	114
Surp_Inf	−0.021	0.000	0.114	0.024	1.200	−0.400	0.300	114
Dif_US_1Y	−0.000	0.000	0.018	1.701	11.132	−0.050	0.110	114
Ret_Gold	0.001	0.000	0.009	−0.190	1.667	−0.029	0.028	114
Ret_BTC	−0.006	0.000	0.101	−4.892	43.346	−0.849	0.288	114
				Panel C: After BTC structural break (2014m1 to 2023m1)				
Ret_SP500	0.000	0.001	0.012	−0.441	6.968	−0.050	0.054	109
Dif_VIX	−0.230	−0.300	1.757	0.876	3.839	−5.710	6.600	109
Surp_Inf	0.006	0.000	0.138	0.925	3.378	−0.400	0.600	109

(*continued on next page*)

Table D1 (*continued*)

Variable	Mean	p50	SD	Asymmetry	**Kurtosis**	Min	Max	Obs.
Dif_US_1Y	0.005	0.000	0.053	2.199	9.413	−0.160	0.230	109
Ret_Gold	0.002	0.003	0.009	−0.289	1.938	−0.029	0.028	109
Ret_BTC	−0.010	−0.000	0.095	−6.607	58.196	−0.849	0.104	109

Note: This table shows statistics for all series covering different sampling periods. The Ret (Dif) prefix indicates that the series was derived from returns (differences) calculated within one-day windows around monthly CPI MoM announcements.

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